

CALDWELL, TX, USA

WMO: 723761

Lat: 30.516N

Lon: 96.704W

Elev: 119

StdP: 99.90

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 23901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.4	(c) -0.9	(d) -10.1	(e) 1.6	(f) 2.9	(g) -7.6	(h) 2.0	(i) 5.4	(j) 10.6	(k) 11.1	(l) 9.3	(m) 12.7	(n) 2.1	(o) 340	(p) 0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.9	(c) 37.4	(d) 22.9	(e) 36.3	(f) 23.2	(g) 35.1	(h) 23.5	(i) 26.1	(j) 29.7	(k) 25.7	(l) 29.9	(m) 25.3	(n) 30.0	(o) 3.6	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	27.0	24.8	20.1	26.8	24.1	19.2	26.5	81.3	29.4	79.6	30.8	78.2	29.6	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.2	7.4	-5.4	38.6	2.8	1.1	-7.4	39.4	-9.0	40.0	-10.6	40.7	-12.6	41.5		
			-6.7	26.4	2.5	0.7	-8.5	26.9	-10.0	27.3	-11.4	27.6	-13.2	28.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	9.2	11.6	16.0	19.6	23.8	27.6	28.6	29.2	26.4	20.6	15.1	10.9
	DBStd	8.10	5.08	5.69	5.02	4.11	3.25	1.98	1.69	1.71	2.74	4.45	5.37	5.27
	HDD10.0	204	75	46	12	1	0	0	0	0	0	1	16	54
	HDD18.3	1016	284	196	104	34	4	0	0	0	1	29	126	237
	CDD10.0	3827	52	89	197	289	427	528	577	596	492	330	167	82
	CDD18.3	1596	2	7	31	72	173	278	319	338	243	101	27	7
	CDH23.3	16696	7	23	143	514	1512	3111	3691	4119	2476	912	165	23
	CDH26.7	7939	0	1	14	116	556	1532	1903	2262	1218	312	25	1
Wind	WSAvg	3.0	3.1	3.5	3.7	3.7	3.3	3.0	2.7	2.4	2.2	2.6	2.9	3.1
Precipitation	PrecAvg	995	85	80	84	83	88	93	54	61	81	113	88	92
	PrecMax	1468	311	233	201	283	167	352	145	161	209	353	243	261
	PrecMin	600	5	5	30	17	0	0	0	5	9	0	11	24
	PrecStd	229	64	59	40	57	53	79	41	46	56	90	65	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	26.0	28.0	32.3	33.9	37.7	37.8	38.8	37.5	33.6	28.9	26.1
		MCWB	17.3	17.6	19.8	21.4	23.1	22.5	23.3	22.5	22.1	22.4	21.6	19.6
	2%	DB	22.2	23.8	26.6	29.1	32.6	36.1	37.1	37.5	36.1	31.7	27.3	23.0
		MCWB	17.0	17.9	19.2	20.5	23.0	22.8	23.1	23.2	22.5	22.2	20.4	19.1
	5%	DB	20.0	22.3	25.0	27.7	31.4	34.1	36.0	36.4	34.0	30.0	25.2	22.2
		MCWB	16.1	17.8	18.9	20.4	22.8	23.2	23.3	23.5	22.5	22.1	19.5	18.3
	10%	DB	17.9	20.9	23.0	26.4	30.0	32.9	34.0	35.5	32.7	28.0	23.1	20.2
		MCWB	14.2	17.2	18.1	19.7	22.4	23.8	23.5	23.8	22.5	20.3	19.3	17.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	20.9	22.3	24.0	25.6	26.8	26.5	26.4	25.9	25.4	23.6	22.1
		MCDB	22.2	22.9	25.7	26.7	30.1	30.4	29.6	29.8	29.3	28.9	26.1	23.5
	2%	WB	18.8	20.1	21.3	23.0	24.8	26.1	26.0	25.9	25.3	24.5	22.4	20.6
		MCDB	20.8	22.3	24.0	27.0	29.3	29.8	29.7	30.1	29.7	28.1	24.8	22.5
	5%	WB	17.3	19.1	20.7	22.4	24.2	25.5	25.6	25.5	24.9	23.8	21.4	19.4
		MCDB	18.5	21.2	23.1	25.8	28.2	29.8	29.9	30.3	29.3	27.2	23.7	21.3
	10%	WB	15.4	17.8	19.7	21.7	23.7	25.0	25.2	25.1	24.4	23.0	20.3	17.9
		MCDB	17.5	19.3	22.4	24.8	27.5	29.5	29.8	30.3	28.8	26.0	23.1	19.2
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.5	11.2	11.4	10.4	10.8	10.9	11.9	11.6	12.4	11.8	11.0
		MCDBR	13.1	11.7	11.1	11.7	11.4	12.3	13.1	13.1	13.4	12.7	12.4	12.6
		MCWBR	8.7	7.7	6.0	5.5	3.9	3.0	2.9	2.8	3.7	4.9	7.0	8.7
	5% WB	MCDBR	10.8	9.4	9.3	9.4	9.2	9.4	10.2	11.4	10.8	10.5	10.2	10.8
		MCWBR	9.3	7.7	6.4	5.3	4.2	3.4	3.1	3.0	3.5	5.2	7.1	9.0
Clear-Sky Solar Irradiance	taub		0.324	0.341	0.364	0.410	0.456	0.442	0.458	0.448	0.418	0.374	0.367	0.324
	taud		2.503	2.469	2.405	2.291	2.207	2.298	2.283	2.304	2.340	2.445	2.402	2.513
	Ebn at Noon		909	923	920	884	841	849	833	839	854	874	848	890
	Edh at Noon		89	100	114	133	146	132	134	130	121	102	98	84
All-Sky Solar Radiation	RadAvg		2.92	3.31	4.43	5.37	5.78	6.45	6.53	6.16	5.16	4.26	3.17	2.56
	RadStd		0.29	0.62	0.41	0.41	0.45	0.47	0.49	0.34	0.47	0.57	0.35	0.32

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86

Nomenclature: See separate page